

PRODUCTS

- Air Cooled Vibration Systems
- DSA Power Amplifiers
- Vibration Controllers
- Bump/Shock Test Machine
- Vibration Application for Elastomers & Transmission Line Dampers

SERVICES

- Engineer & Customer Training
- Maintenance Contract Services
- Refurbished System Supply
- Spare Part Supply
- Armature Coil Rewinding
- Replacement of Power Amplifiers
- Application Support

TYPICAL SYSTEM APPLICATIONS

- Avionic Testing
- Automotive Component Testing
- Electronic Component/Assembly Testing
- Stress Screening
- Sine, Random & Transient Testing
- Natural Frequency Search
- Life Time Simulation
- Product Design Validation
- Product Qualification
- Transport & Packaging
- Halt/HASS Simulation



SYSTEM RATINGS

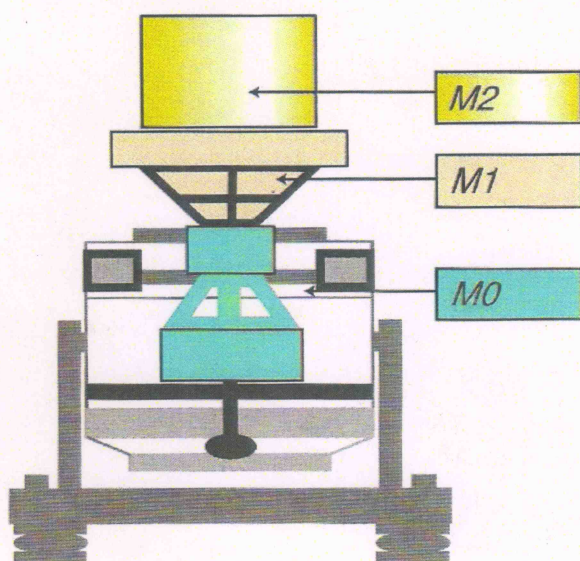
Electro Dynamic Vibration Test Systems are used to satisfy a wide variety of vibration test requirement. Sine, Random and Shock testing are considered the three basic test environments and are reflected in the standard ratings for all Apson vibration test systems. All vibration test systems are rated in peak force for sine and shock and in rms force for random. All systems are rated to 2,000 Hz. Some systems will deliver full performance to 3,000 Hz. All systems will operate to 3,000 Hz or beyond at reduced performance.

SINE : Sine Performance is defined by an envelope of maximum displacement, velocity, and acceleration as a function of frequency for a given Amplifier and shaker combination.

RANDOM : Random performance is based upon a grms level over a specific frequency range. In order to ensure uniformity a specific set of conditions must be used to eliminate test variables. These conditions include a flat spectral density (g^2 / Hz) from 20 Hz to 2,000Hz and a heavy, non-resonant load. This method of rating provides consistency and exceeds the requirements of ISO 5344, if a shaped spectrum is used, the random force rating may vary.

SHOCK : Shock performance is defined for various pulse waveforms is half sine, saw tooth. A family of curves is provided for each pulse waveform based upon a given pulse duration and peak acceleration.

RATED FOREC CALCULATION :



CALCULATION :

1.	SINE FORCE (p-p) $F = M \times A = (M0 + M1 + M2) \times a \times 1.2$
	M0 : Armature mass (kgw)
	M1 : Auxiliary Table or Fixture (kgw)
	M2 : Specimen (kgw)
	a : Vibration acceleration (g)
	1.2 : Safety Factor
2.	RANDOM FORCE (RMS) $= \text{SINE FORCE} \times \sqrt{2}$

$$\Sigma M = M_0 + M_1 + M_2$$

Note : Because of continuous product improvement, specifications are subject to change without advance notice. Latest status will always be indicated in our quotation.



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